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A Lightweight Traffic Light Detection and Recognition Model Based on YOLOv5

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***Abstract*—With the rapid development of intelligent mobile platforms, real-time and accurate Traffic Light Detection and Recognition (TLDR) is critical for ensuring traffic safety. However, existing TLDR methods fail to balance detection accuracy, speed, and deployment feasibility. To address this, we propose a high-precision lightweight TLDR model. First, an enhanced backbone network based on the EfficientViT architecture is proposed, incorporating an efficient multi-scale attention module to extract rich semantic features while reducing computational load. Second, a lightweight cross-scale feature fusion network is constructed to enhance information interaction between deep and shallow features. Finally, the model is trained and validated on the open-source S2TLD dataset. Experimental results demonstrate that the proposed model achieves a detection speed of 204 FPS, precision of 95.0%, and 2.1 GFLOPs. Compared to YOLOv5n, the proposed model improves detection speed by 35.1%, precision by 4.7%, and GFLOPs by 48.8%, respectively. This work provides technical support for constructing lightweight TLDR models and offers theoretical reference for deploying TLDR models on intelligent mobile platforms.**

Keywords - Traffic light detection and recognition; lightweight network; deep learning; improved YOLOv5; intelligent mobile platforms

I. INTRODUCTION

As pivotal components of transportation systems, traffic lights play a vital role in regulating right-of-way allocation. Real-time and accurate TLDR is essential for assisting mobile platforms in planning driving strategies and preventing accidents caused by signal misjudgments. However, TLDR in real-world scenarios face challenges such as complex background interference, multi-scale object distribution, and extreme illumination variations. Existing TLDR methods include handcrafted feature-based traditional methods and deep learning-based methods. Diaz-Cabrera et al. [1] used the fuzzy clustering and sequence rules to detect traffic lights according to color features. Chen et al. [2] located the regions according to the characteristics of the traffic light boards, and then used the color information to recognize the traffic lights. Joo et al. [3] proposed a detection algorithm based on video, which is composed of rotating principal component analysis and amplitude threshold of PC plane histogram. These traditional methods rely on the color, shape and other unique characteristics of traffic lights, but it is difficult to ensure good detection performance in complex scenarios.

While deep learning has advanced object detection considerably, its direct application to traffic light recognition remains challenging. Currently, deep learning-based TLDR methods primarily adopt either two-stage region-based frameworks or one-stage regression approaches. Hassan et al. [4] proposed a deep learning-based TLDR system, which incorporated the specific shape, color and texture of traffic lights into the network learning process. Wang et al. [5] proposed a two-stage detection model combining low-light image enhancement and small target detection, which has high detection precision under low-light conditions. Niu and Li [6] proposed a TLDR model based on YOLOv5s target detection and AlexNet image classification. Li et al. [7] used the aspect ratio, area, location and context of traffic lights as prior information, and used the aggregated channel feature method and inter frame information analysis method to achieve high-precision TLDR. Wang et al. [8] proposed a two-stage TLDR model, which uses improved YOLOv5s and extended twin support vector machine to detect and recognize traffic lights. Kim et al. [9] proposed a two-stage method, which is composed of a pixel-wise semantic segmentation technology and a full convolution network. Bach et al. [10] developed a TLDR system based on the Faster R-CNN, which can not only detect and recognize traffic lights, but also distinguish their shapes. Shi et al. [11] proposed an adaptive background suppression algorithm to highlight the candidate regions. Ouyang et al. [12] proposed a lightweight model for TLDR. The model consists of a heuristic candidate region selection module and a lightweight convolutional network classifier.

Although the two-stage target detection methods have achieved high detection accuracy by step-by-step detection

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mechanism, its complex multi-level architecture leads to high computational cost. Therefore, the two-stage methods are difficult to achieve efficient deployment in the edge computing platforms. In contrast, the one-stage methods rely on end-to-end learning mechanism to significantly reduce the complexity of the model, which is more suitable for mobile platforms. Muller and Dietmayer [13] used SSD algorithm to detect traffic lights and achieved a detection speed of 10 FPS on NVIDIA Titan XP GPU. Yao et al. [14] proposed a lightweight model TL-detector to achieve a mAP of 73.24% on the TL2022 dataset. Ouyang et al. [15] proposed a multi-backbone network. The network consists of three convolution backbones, i.e., normal, residual and highway convolution modules, and uses channel compression to control the model parameters. Li et al. [16] improved the YOLOv5 algorithm and achieved high accuracy on the BDD100K dataset. Lai et al. [17] proposed a small target detection model STC-YOLO, and enhanced the TT100K dataset to improve the anti-interference ability of the model. Chuang et al. [18] proposed a target detection model for traffic light classification at a distance. The model consists of a parallel residual bi-fusion feature pyramid network and attention modules. Liu et al. [19] proposed a symmetrical traffic sign detection algorithm. Although this algorithm is superior to the YOLOv5l in detection performance, it is still limited by the huge computational load. Behrendt et al. [20] constructed the Bosch small traffic light dataset, which contains 5000 images and a video sequence of 8334 frames. Lee and Kim [21] proposed a deep neural network for real-time TLDR. The research shows that the network has good detection performance in Bosch small traffic light dataset and LISA traffic light dataset. Wang et al. [22] improved YOLOv4 by using feature enhancement and bounding box uncertainty prediction mechanism. The results show that the average precision of the improved algorithm on the LISA dataset is improved by 2.86%. In order to improve the recognition accuracy, John et al. [23] used the on-board GPS sensor to identify the region-of-interest containing traffic lights. Yoneda et al. [24] proposed a TLDR algorithm using the digital map and precise vehicle pose. Lu et al. [25] proposed a illumination invariant model based on YOLO, which has excellent detection performance in traffic scenes with different illumination. Liu and Lin [26] designed the dark channel de-fogging algorithm, which effectively improved the detection performance in the fog environment. Zhang and Hu [27] proposed a YOLOv7-based multi-class target detection method, which can accurately detect multiple types of traffic targets in foggy weather. Although compared with the two-stage methods, the one-stage methods have cheaper computational cost and faster detection speed. However, the existing one-stage methods still fail to achieve real-time detection on the mobile platform.

With the emergence of intelligent mobile platforms, the mobile devices such as autonomous vehicles and low-altitude drones are gradually integrated into the modern transportation system. In this background, real-time and accurate TLDR has become the core technical requirement to ensure the traffic safety of intelligent mobile platform. Noted that due to the computational power, power consumption, weight and other limitations of edge computing platforms, the existing TLDR models developed based on high-performance computing architecture are difficult to be widely and directly deployed on these platforms. Therefore, a one-stage traffic light detection model for mobile platform is proposed, which can reduce the computational cost and ensure high detection accuracy, so as to improve the feasibility of deployment on mobile platform. The improvement measures and main contributions of this paper are summarized as follows:

(1) An enhanced backbone network is proposed. The network is based on EfficientViT architecture, and introduces an efficient multi-scale attention mechanism based on the dual dimensions of space and channel to promote the original network to understand the spatial feature dependence, and extract rich small target location information, so as to improve the detection performance of the model for small-scale traffic lights.

(2) A lightweight neck network based on YOLOv5 is constructed. Firstly, we construct a lightweight convolution process and combine it with the traditional convolution modules CBS and C3 to reduce the computational cost without changing the output feature map. In addition, a cross-scale feature fusion network is designed to realize the feature fusion between shallow information and deep information, and further improve the learning ability of the model for small target features.

The rest of this paper is divided into the following parts: In Section 2, the structure of the model and the corresponding improvements are introduced; In Section 3, the platform and dataset used are shown; In Section 4, the evaluation and validation process and results of the model are described; In Section 5, the conclusion is given.

II. METHODOLOGY

Compared with the YOLO series [28], [29], [30], [31], YOLOv5n algorithm has the least parameters. Therefore, we use YOLOv5n as a benchmark and further improve it to develop a TLDR model suitable for mobile devices.

A. Enhanced backbone network

The backbone network provides rich semantic information for subsequent networks. However, to extract deep semantic information, the traditional backbone network is usually too complex to retain the details of small targets such as traffic lights. At the same time, this complexity will generate substantial computing costs, which is difficult to meet the real-time requirements of the edge computing platform. To solve this problem, EfficientViT [32] is used as the basic backbone network. The network provides a lightweight architecture for TLDR by integrating deep separable convolution and multi-scale linear attention. However, the channel-based attention mechanism exhibits limitations in capturing cross-scale spatial feature, particularly for low-resolution targets [33]. To this end, an efficient multi-scale attention mechanism is introduced, and its structure is shown in Figure 1. This module adopts the feature grouping strategy to achieve dual-dimensional decoupling of spatial and channel features, and uses the spatial feature dependency to extract rich location features. Notably, the module enhances Detail features while reducing computational load.

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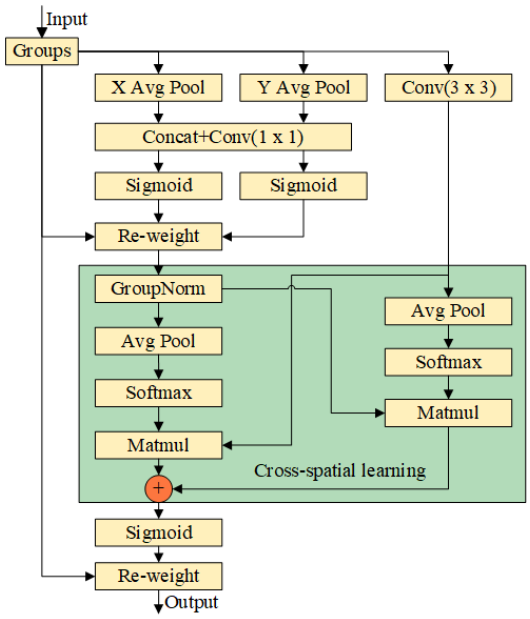


Figure 1. efficient multi-scale attention module structure

B. Lightweight neck network

1) g module

To integrate the feature information and strengthen the learning ability, the neck network is often consists of many convolution layers. However, this leads to a lot of computing costs, which is not conducive to building a lightweight model [34]. Therefore, in order to reduce the computational cost without reducing the ability of feature fusion, we constructed a high-efficiency and lightweight convolution operation g module. Compared to the CBS module, the g module can obtain feature map of the same scale with lower computational cost. As shown in Figure 2, assume that the feature X' is calculated by the traditional convolution module CBS, and then perform linear operation on the feature X' to finally obtain the feature map Y. Compared with the traditional convolution operation, this process not only reduces the number of parameters, but also can get the same size feature map. Therefore, we replace the traditional convolution module CBS in the neck network with the g module, so as to reduce the computational cost as much as possible without losing the model's detection performance.

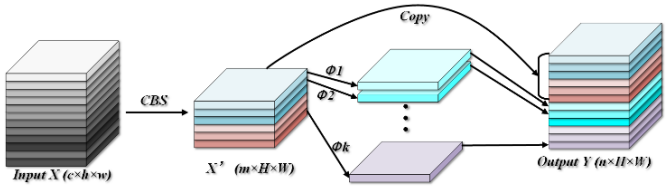


Figure 2. g module structure

The g module consists of CBS module and linear operation $\phi=[\phi_1,\phi_2,...,\phi_k]$. Assume that the input feature X is $c\times h\times w$, where c represents the number of channels, h and w represents the height and weight of the feature map, respectively. And the CBS module is $m\times k\times k$, where m is the number of channels of the CBS module, k is the size of the CBS module. Thus, the number of parameters of the CBS is $m\times k\times k\times H\times W\times c=n/s\times k\times k\times H\times W\times c$, where s represents proportionality factor, i.e., $s=n/m$. Subsequently, the feature X' through the linear operation ϕ. Finally, X' is concatenated with the feature map after linear operation to output feature map $Y(n\times H\times W)$. Therefore, the number of parameters of linear operation is $H\times W\times m\times d\times d\times(s-1)=H\times W\times n/s\times d\times d\times(s-1)$. However, for the traditional convolution operation, assuming that the input feature and output feature are still X and Y, respectively, the number of parameters of the convolution is $n\times H\times W\times k\times k\times c$. Thus, parameter ratio γ of traditional convolution module and g module can be expressed as:

$$\gamma=\frac{n\times H\times W\times k\times k\times c}{n/s\times k\times k\times H\times W\times c+H\times W\times n/s\times d\times d\times(s-1)}=\frac{k\times k\times c}{k\times k\times c/s+d\times d\times(s-1)/s} \tag{1}$$

where $d=k$ and $s\ll c$. Thus, parameter ratio γ can be calculated as:

$$\gamma=\frac{k\times k\times c}{k\times k\times c/s+d\times d\times(s-1)/s}=\frac{c\cdot s}{c+s-1}\approx s \tag{2}$$

Therefore, the traditional convolution requires s times as many parameters as the g module. Therefore, compared with the traditional convolution, the g module can generate feature map of the same size, while significantly reducing the computational cost.

In addition, C3 module is also the core module of YOLOv5, and this module is composed of multiple traditional convolution layers. Therefore, the C3 module will also generate a high computational load. To this end, we combined C3 module with g module to construct C3G module. The specific structure is shown in Figure 3. The C3G module uses the cheap linear operation of g module to generate rich feature information. While replacing the traditional bottleneck module, C3G module retains the multi-branch feature fusion ability of C3 module, and effectively enhances the robustness to small-scale traffic lights and complex scenes.

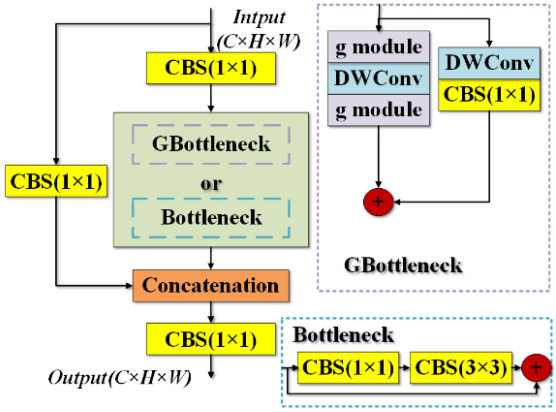


Figure 3. C3 and C3G module structure

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1) C-PAN

With the development of deep learning technology, network architectures such as FPN, PAN and BIFPN have gradually promoted the optimization of multi-scale feature fusion. FPN integrates deep features with shallow features through a top-down path, so as to build a multi-scale feature pyramid. PAN adds a bottom-up enhancement path compared with FPN to form bidirectional feature fusion. BIFPN further optimizes the path design, dynamically adjusts the weights of different levels of features through learnable weight parameters, and introduces cross-scale jump connection to further promote feature fusion. Although PAN has become the core component of lightweight detection model with its lightweight architecture, it still has the problem of insufficient multi-scale feature fusion ability for small target detection tasks such as TLDR. Inspired by BIFPN [35], a cross-scale feature fusion network C-PAN is proposed. The network promotes the aggregation of multi-scale feature information, which improve the ability of feature extraction with fewer parameters. Its structure is shown in Figure 4.

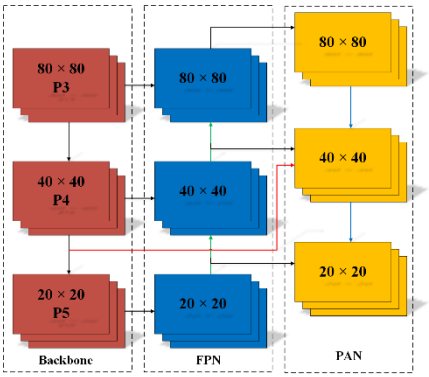


Figure 4. C-PAN network structure

As shown in Figure 4, the enhanced backbone network extracts three feature maps of different sizes from the original image, namely P3, P4 and P5. Compared with FPN+PAN structure, the C-PAN structure adds an information channel between deep features and shallow features. This structure can effectively integrate shallow detail features and deep semantic features to avoid the disappearance of small target features with the deeper of the network, thus enhancing the learning ability of the model for traffic light images with fewer pixels.

Through the improvement measures described in Section 2, a lightweight TLDR model is proposed, and its structure is shown in Figure 5. Firstly, the input features are reshaped to 640 × 640 feature maps. Secondly, the model gradually down-sampling the original image to extract P3, P4, P5 feature maps through the enhanced backbone network. Finally, the proposed model fuses the features extracted from the backbone network through the C-PAN network, and uses the YOLOv5 detection head network to detect the feature maps of different scales.

III. PLATFORM AND DATASET

The high computing-power NVIDIA GPU platform is used as the training platform. Table 1 describes the Specification of the training platform. The platform is equipped with NVIDIA RTX 3060 12GB GPU.

Table 1 Specification of the training platform

Attribute	Value
CPU	Intel Core i5 13400
GPU	NVIDIA RTX 3060
Random Access Memory (GB)	32
Extend Computing Power (TFLOPS)	102
Operation System	Windows10

The dataset used in this study is the S2TLD (720x1280) dataset collected in the real scene [36]. The dataset contains scenes with different illumination conditions, such as night, exposure. Figure 6 is an example image of the S2TLD (720x1280) dataset. This dataset comprises 4,564 annotated traffic images: 3,785 samples for training and 779 for validation. And training/validation sets containing 5,712/761 red, 155/35 yellow, and 3,455/1,073 green light instances, respectively.



Figure 6. Example images of the S2TLD dataset

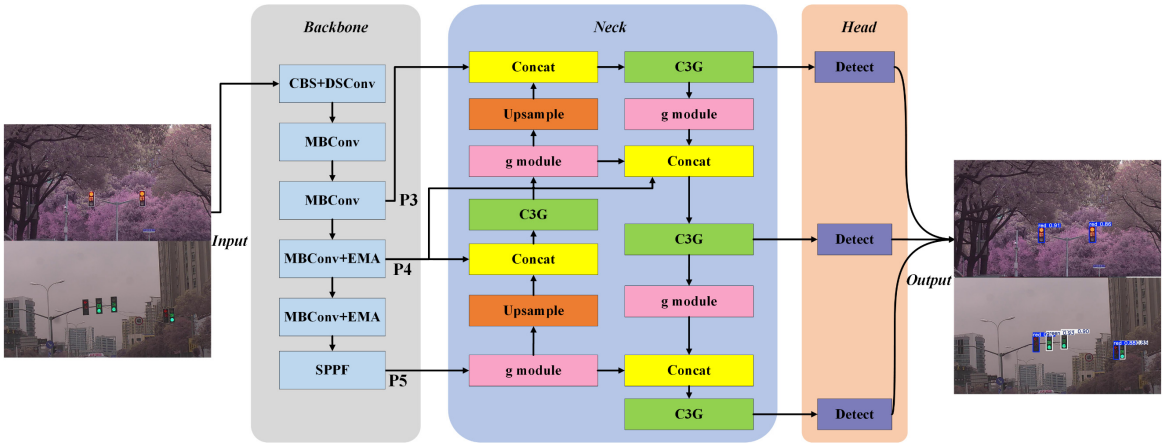


Figure 5. Overall model structure

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IV. RESULTS

In this section, the model performance evaluation indicators are introduced and the detection performance of the model is demonstrated. Finally, the ablation experiment is described and the comparison results are shown.

A. evaluation indicators

The Precision (P), Recall (R), mAP, F1-score, size, and GFLOPs are utilized to evaluate the detection performance. The P can be defined as:

$$P = \frac{TP}{TP + FP} \tag{3}$$

where TP stands for True Positive and FP stands for False Positive. The R can be expressed as:

$$R = \frac{TP}{TP + FN} \tag{4}$$

where FN stands for false negative. The F1-score is a comprehensive indicator of P and R, which is defined as follows:

$$F1 - score = 2 \times \frac{P \times R}{P + R} \tag{5}$$

The AP represents the average precision of a certain category, the mAP represents the average precision of all categories, and they can be defined as:

$$AP = \int_0^1 P dR \tag{6}$$

$$mAP = \frac{1}{n} \sum_{i=1}^{n} AP \tag{7}$$

B. Detection effect of proposed model

YOLOv5 has five versions of different sizes, x, l, m, s and n in descending order. Considering the hardware limitations, it is difficult for the mobile platform to run the high-performance YOLOv5 version in real time. Therefore, the detection performance of the proposed model is compared with that of YOLOv5s and YOLOv5n. Figure 7 shows the heatmaps of the detection effect of each model on some samples, and figure 8 shows the TLDR effect of different models on some samples. Table 2 and Figure 9 show the detection performance and confusion matrix of the model, respectively.

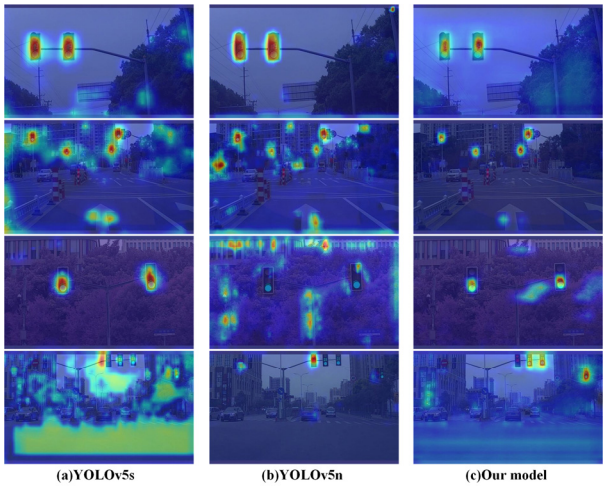


Figure 7. Heatmap of detection effect for different models (Partial Samples)



Figure 8. TLDR effect of different models (Partial Samples)

Table 2 Detection performance of proposed model

Attribute	Value
Precision (%)	95.0
Recall (%)	90.2
F1-score (%)	92.5
mAP (%)	95.5

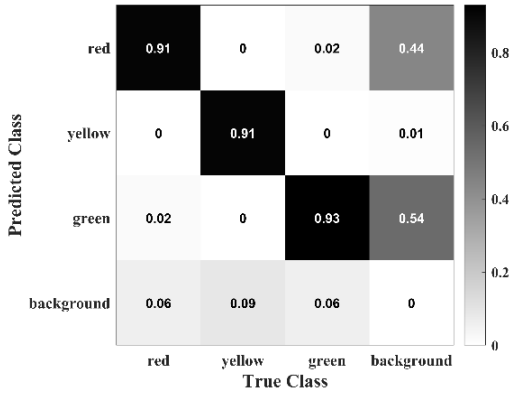


Figure 9. Confusion matrix of proposed model

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It is obvious from Figure 7 that the attention of the proposed model is mainly focused on the traffic lights. However, the attention of other models is relatively scattered and vulnerable to environmental interference. The detection results in Figure 8 show that our model has good detection effects under different illuminations. Table 2 and Figure 9 show that the proposed model has excellent detection performance, and the detection precision of traffic lights can reach 95%.

C. Ablation study

To systematically assess the impact of architectural improvements, five ablation experiments were designed where all baseline configurations were initialized with random weights. Table 3 shows the experimental configuration.

Table 3 Setting of experimental parameters

Attribute	Epoch	Patience	Batch size	Optimizer
Value	300	100	16	Adam

Table 4 Results of ablation experiments

Experiments Number	P (%)	R (%)	F1-score (%)	GFLOPs	Parameters (Million)	Model Size(MB)
1	90.3	90.0	90.1	4.1	1.8	3.6
2	89.9	86.3	88.1	3.0	1.5	3.1
3	91.3	92.0	91.6	2.7	1.2	2.6
4	96.2	88.0	91.9	2.7	1.3	2.7
5	95.0	90.2	92.5	2.1	1.1	3.1

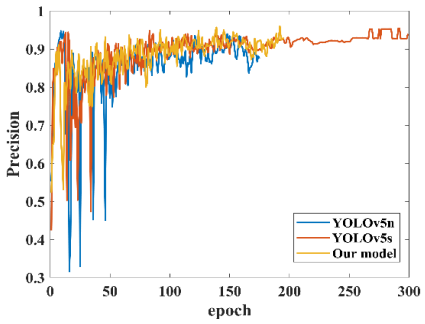
In the ablation experiment, experiment 1 directly uses the YOLOv5n to recognize traffic lights. In Experiment 2, traffic lights were detected using a model composed of EfficientViT+YOLOv5 network. On the basis of Experiment 2, Experiment 3 further introduced efficient multi-scale attention mechanism. Experiment 4 is an improved model constructed by using enhanced backbone network and C-PAN feature fusion network. Experiment 5 is a lightweight improvement on CBS and C3 based on Experiment 4. The results of ablation experiment are shown in Table 4.

According to the results in Table 4, the proposed model architecture leads to marked improvements in detection performance. Specifically, the calculation load, number of parameters and size of the model are 2.1GFLOPs, 1.1M and 3.1MB, respectively. Compared with the YOLOv5n, the amount of calculation, parameters and model size of the model are reduced by 48.8%, 38.9% and 13.9%, respectively. In addition, the proposed model achieves a precision of 95%, a F1-score of 92.5%. Compared with the original YOLOv5n model, it increased by 4.7% and 2.4%, respectively.

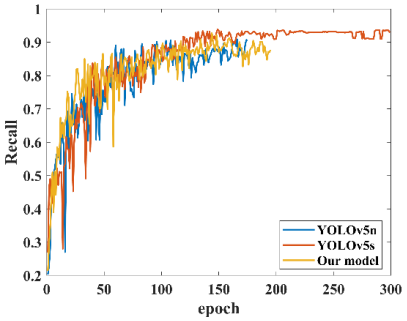
By comparing the experimental results of 1, 2 and 3, it can be seen that the improvements in Section 2.1 have effectively reduced the number of parameters and GFLOPs by 34.1% and 33.3%, respectively. At the same time, the improvements also improved the precision by 1.0% and the recall by 2.0%. In addition, according to the results of experiments 4 and 5, it can be seen that the precision can be improved by 3.7% with C-PAN network and ghost module.

D. Comparison of Detection Performance

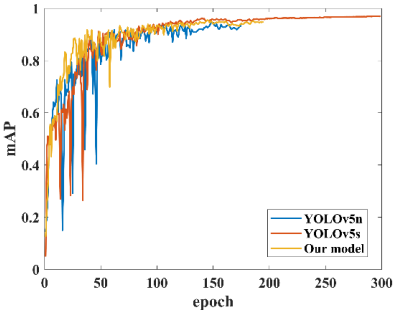
Figure 10 (a) (b) (c) shows the comparison of P, R, mAP curves between the proposed model and the YOLOv5n, YOLOv5s models during the training process. It can be seen that the fluctuation amplitude of each index of the proposed model is the smallest in the training process, while the fluctuation of YOLOv5n and YOLOv5s models is larger in the training process, especially in the P. In addition, the performance improvement speed of the model built at the beginning of training is higher than that of the YOLOv5n and YOLOv5s models, and the precision of the model built is also higher than that of the YOLOv5n and YOLOv5s models. Note that although the proposed model and YOLOv5n can quickly reach a stable state, YOLOv5n is lower than the proposed model in all aspects.



(a) Precision Comparison Curve



(b) Recall Comparison Curve



(c) mAP Comparison Curve

Figure 10. Comparison curve of detection performance

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Table 5 Results of model comparison

Model Type	Model Size (MB)	GFLOPs	Parameters (M)	P (%)	R (%)	F1-score (%)	mAP (%)	FPS
YOLOv5s	13.6	15.8	7.0	92.0	**93.2**	**92.6**	**96.6**	161
YOLOv5n	3.6	4.1	1.8	90.3	90.0	90.1	95.2	151
YOLOv8n	6.3	8.1	3.0	89.5	86.7	88.1	91.3	152
YOLOv10n	5.5	8.2	2.7	91.8	89.9	90.8	94.9	167
Our model	**3.1**	**2.1**	**1.1**	**95.0**	90.2	92.5	95.5	**204**

Table 5 shows that the comparison results of different models in size, GFLOPs, parameter, precision, recall, F1-score, mAP and FPS. The results show that the running speed is 204 FPS, and the precision is 95.0%. Compared with other models, the maximum improvements are 35.1% and 5.5%, respectively, which are higher than other models, and the calculation cost is lower than other models. In addition, compared with other YOLO series, the proposed model is second only to YOLOv5s in mAP and F1-score, reaching 95.5% and 92.5%, respectively.

V. CONCLUSION

In the urban intelligent transportation system, the accurate and real-time TLDR plays a key role in ensuring the traffic safety of intelligent platforms such as autonomous vehicles and mobile devices. However, due to the limitations of computing power, power consumption and cost of mobile platforms, it is difficult to widely deploy high-performance TLDR models on mobile platforms. Therefore, a lightweight TLDR model is proposed to achieve real-time and accurate traffic light detection on the mobile platform. Firstly, an efficient multi-scale attention mechanism based on channel and space is introduced into EfficientViT to enhance the attention of the backbone network to the location information of traffic lights, thus enhancing the performance of the model to small targets. Secondly, the g module, C3G module and C-PAN network are constructed in the feature fusion network to promote the interaction between deep features and shallow features, and reduce the computational load without changing the scale of the output feature. Finally, it is verified on the S2TLD dataset. The results show that the precision, recall and F1-score of the model are 95.0%, 90.2% and 92.5%, respectively. At the same time, the model has only 1.1 M parameters, 2.1 GFLOPs and 3.1 MB.

This study can provide technical support for lightweight TLDR model and small target detection method, and provide theoretical reference for model deployment on intelligent vehicles, mobile robots and other devices. In the next research work, we will focus on the problem of traffic light detection under extreme weather conditions, especially in low-visibility environments such as rain and fog.

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